BSIT 375: Knowledge Discovery and Data Science

Comprehensive Module: Cloud-Scale Data Engineering and Machine Learning with AWS

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1. Introduction: The Modern Data Lakehouse Architecture

In traditional Data Science, we often worked with static CSV files on local machines. However, real-world knowledge discovery requires processing massive, high-velocity data. AWS provides a **Medallion Architecture** (Bronze, Silver, Gold layers) to manage this flow.

> **The Data Flow:** Ingestion (Kinesis/SQS) → Raw Storage (S3 Bronze) → ETL/Processing (Glue/Spark) → Cleaned Storage (S3 Silver) → Analytics (Athena/Redshift) → Intelligence (SageMaker).

2. Data Ingestion & Streaming

2.1 Amazon Kinesis

Amazon Kinesis is used for real-time data streaming. In your Wikipedia project, Kinesis acts as the entry point for "live" edits. It allows us to process data as it arrives rather than waiting for batch uploads.

- **Data Streams:** Low-latency streaming for custom applications.
- **Data Firehose:** Automatically loads streaming data into S3 or Redshift.

2.2 Amazon SQS (Simple Queue Service)

A fully managed message queuing service. SQS is vital for **decoupling** systems. If our Spark job is slow, SQS holds the incoming notifications so no data is lost during spikes in Wikipedia activity.

3. Cloud Storage: Amazon S3 (Simple Storage Service)

S3 is the backbone of the Data Lake. It is an object storage service offering industry-leading scalability and data availability.

Layer	Name	Purpose
Bronze	Raw Edits	Original JSON/CSV data from Wikipedia API.
Silver	Processed Edits	Cleaned data, types casted, bots filtered (The *MyTransform* output).
Gold	Aggregated Data	Ready for dashboards and final mathematical modeling.

4. Data Transformation: AWS Glue & Apache Spark

4.1 AWS Glue

Glue is a serverless data integration service. It provides a **Data Catalog** (a central metadata repository) and ETL (Extract, Transform, Load) capabilities.

4.2 Apache Spark (PySpark)

Spark is a distributed processing engine. In Glue, we use PySpark to manipulate **DataFrames**. As seen in our recent lab, we use Spark functions (`F.abs()`, `F.when()`) to apply mathematical logic across millions of rows simultaneously.

```
# Example PySpark Transformation for Data Science
processed_df = df.withColumn("edit_impact",
    F.when(F.abs(F.col("size")) > 1000, "High").otherwise("Low"))
```

5. Data Analytics & Warehousing

5.1 Amazon Athena

Athena is an interactive query service that makes it easy to analyze data in S3 using standard SQL. It is **serverless**; you only pay for the queries you run. It uses the Glue Data Catalog to understand the schema of your files.

5.2 Amazon Redshift

A petabyte-scale data warehouse. While Athena is for ad-hoc exploration, Redshift is used for complex analytical queries and high-performance reporting. It stores data in a columnar format, making mathematical aggregations (SUM, AVG, VAR) extremely fast.

6. Machine Learning: Amazon SageMaker

SageMaker is a fully managed service that provides every developer and data scientist with the ability to build, train, and deploy machine learning (ML) models quickly.

Mathematical Link: We use SageMaker to apply the ANN concepts discussed in Week 5 (Calculus and Linear Algebra) to our S3 data. SageMaker handles the gradient descent and backpropagation math on cloud GPUs.

7. Summary Table: Service Roles in Data Science

Service	Phase	Data Science Task
Kinesis	Ingestion	Real-time feature collection.
S3	Storage	Data Lake management.
Glue/Spark	ETL	Feature Engineering & Cleaning.
Athena	Analytics	Exploratory Data Analysis (EDA).
SageMaker	ML	Model Training & Deployment.